DEEP RESEARCH AGENT

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Deep Neural Networks
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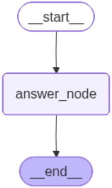
Problem Statement

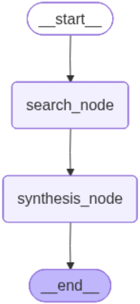
- Large language models are strong tools for natural language understanding and generation. However, when used for research-oriented tasks, they often struggle from limitations like shallow reasoning, made-up facts, absence of citations, weak source grounding, and insuffiecient iterative improvement.
- Traditional chatbot like systems typically generate responses in a single pass. This makes them unfit for complex research questions that needs planning, multi-step reasoning, source comparison, and factual validation.

Project Objective

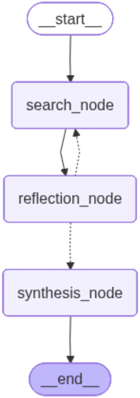
The main objective of this project is to design and build a deep research agent that decomposes a complex question, collects evidence, verifies sources, and synthesizes a final well-written answer.



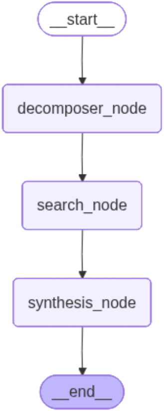
Single LLM Call Architecture



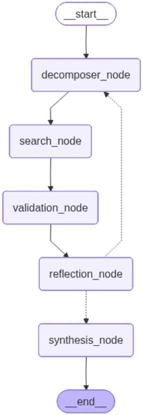
Single Search + LLM Architecture



Iterative Search + LLM Architecture



Decomposer Search Synthesis



Architecture developed in Assignment

Architecture

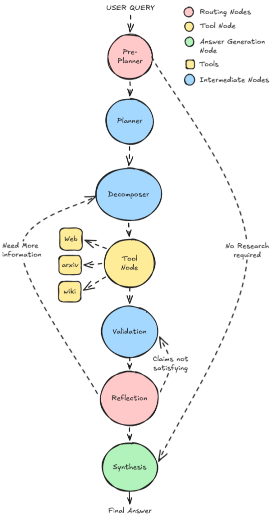
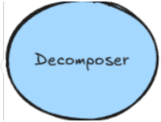


Figure 1: Complete Architecture of the Deep Research Agent

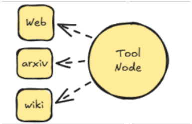
Node-Level Information

Determines whether the user query requires research and acts as the first decision-making node.

Generates a structured research plan based on the query, research depth, and selected tools.



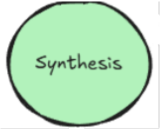
Breaks the research task into smaller focused sub-queries for easier retrieval and synthesis.



Fetch information from external sources like Web, Wikipedia, and Scientific databases.

Extracts and verifies key claims from retrieved sources for reliability.

Checks if the collected information is sufficient or if more research is needed.



Produces the final structured and well-written research response.

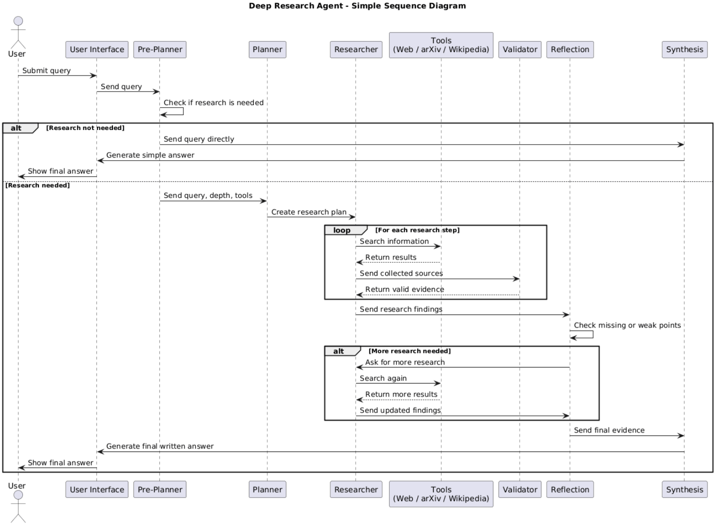


Figure 2: Workflow of the system as a sequence diagram

Technology Stack

Backend Technologies

Frontend Technologies

Frontend UI Overview

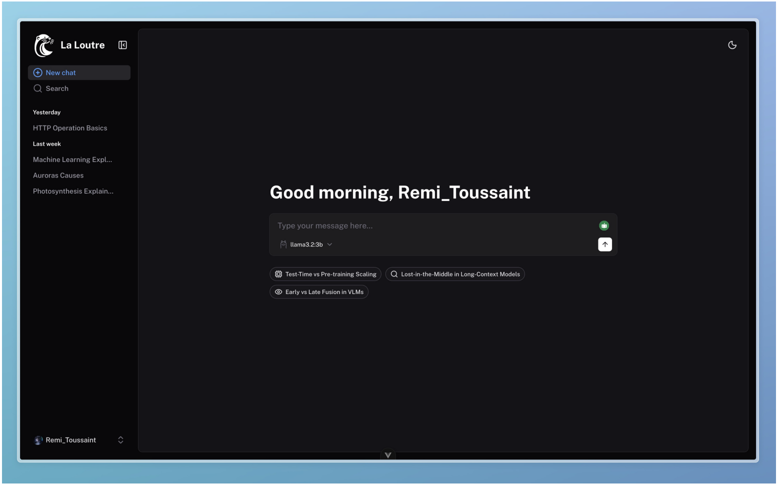


Figure 3: Home page of *La Loutre* in dark mode, with a personalised greeting in English and the three domain-specific research prompts.

Vue 3 + Vite

Chat UI

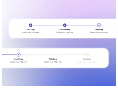
Streaming

Responsive Dashboard

Markdown

Dark Mode

Title generation



UI indicators

Multilingual UI

CSRF

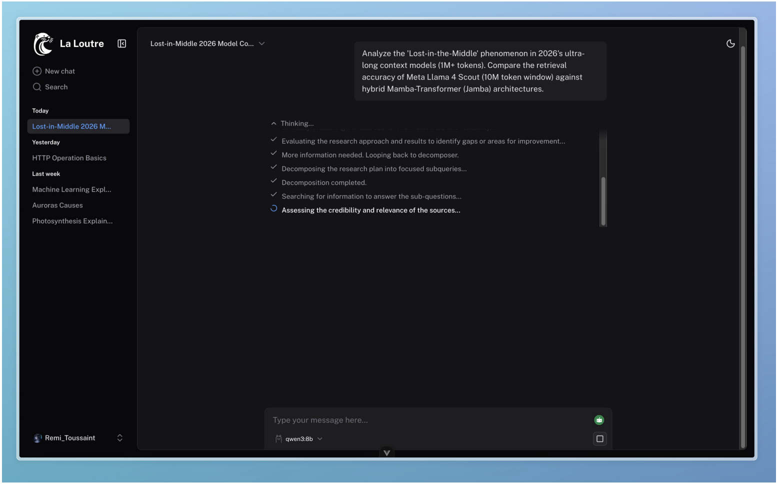


Figure 4: An active research session showing the task-list reasoning component. Completed steps are marked with a check icon, and the active step is indicated by a spinning loader.

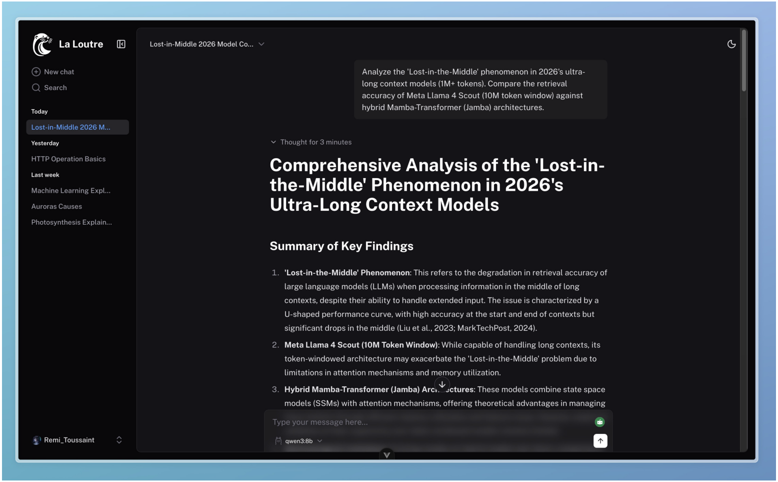


Figure 5: Top of a completed response: collapsed reasoning block shows thinking duration, followed by the opening of the structured final answer.

Results & Contributions

Architecture	Context Relevance	Answer Relevance	Faithfulness / Groundedness	Decomposition Quality	Reflection / Self-Correction
Single LLM Call	0	10	0	0	0
Single Search + LLM	10	10	10	0	0
Iterative Search + LLM	9	10	10	0	0
Decomposer Search Synthesis	9	10	5	9	0
Our Architecture	10	10	9	9	6

- The evaluation is based on a set of research-focused questions designed to test different abilities of the system. These include factual retrieval, scientific understanding, multi-step reasoning, source validation, and the ability to synthesize information into a final answer.
- Each architecture is evaluated using a rubric-based scoring framework powered by an LLM as-a-Judge evaluation system. The resulting scores are then compared across architectures.

Future Works

- Reflection-Driven Tool Expansion
- Human-in-the-Loop After Planning
- Stronger Evaluation Framework
- Improved Source Validation
- Full-Text Research Papers Retrieval
- Adaptive Loop Control
- Parallel Multi-Agent Collaboration

Conclusion

- Developed a Deep Research Agent using an agentic AI workflow
- Overcame limitations of traditional single-prompt LLM systems
- Implemented planning, decomposition, validation, reflection, and synthesis stages
- Enabled human-like multi-step research and evidence gathering
- Improved reliability, completeness, and transparency of generated answers
- Provided a more structured and trustworthy AI-assisted research approach
- Trade-off: increased latency and dependency on external tools

THANK YOU!